Motif finding in DNA sequences using EM algorithm

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Introduction

The Expectation-Maximization (EM) algorithm is a method used to find the most likely model parameters when some data is missing or hidden. It is especially useful in models with hidden variables, such as mixture models, where direct maximization of the likelihood is not possible. The algorithm works by repeating two steps:

1. the Expectation step (E-step), which computes the expected value of the hidden variables given the current parameters,
2. the Maximization step (M-step), which updates the model parameters by maximizing the expected complete-data log-likelihood.

In this project, we apply the EM algorithm to a simplified version of a DNA motif discovery problem. Motif finding involves identifying short, recurring patterns in DNA sequences that may have biological significance. We work with several DNA sequences of equal length, assuming each sequence was generated either by a motif pattern or by a background process.

Each sequence is assumed to come from one of two probability distributions. If it originates from the motif model, then the nucleotides (A, C, G, T) in the sequence are drawn from a position-specific distribution, where the probability of each nucleotide depends on its position in the sequence. If it comes from the background model, the nucleotides are drawn independently from a single position-independent distribution representing general background nucleotide frequencies.

To determine which distribution generated each sequence, we introduce a hidden binary variable indicating whether the sequence was produced by the motif or the background model. These variables are not observed, but we assume that each sequence has a fixed probability α of coming from the motif model. This parameter α is known in the initial part of the project.

The goal is to apply the EM algorithm to accurately estimate the parameters of both distributions based on the observed data.

1 Model Formulation

We are given a set of DNA sequences of equal length $w = 3$, where each nucleotide is one of four letters $\Sigma = \{A, C, G, T\}$. Let $d = |\Sigma|$, so in this case $d = 4$.

1.1 Motif and background distributions

Each observed sequence $x_i = (x_{i1}, x_{i2}, \ldots, x_{iw})$ is assumed to be generated from one of two possible probability distributions:

- **Motif model:** The sequence is generated according to a position-specific distribution over nucleotides. For each position $j = 1, \ldots, w$, we define a distribution

$$\theta_j = (\theta_{1,j}, \ldots, \theta_{d,j})^T,$$

where $d = 4$ corresponds to the four nucleotides $\{A, C, G, T\}$. These probabilities satisfy the condition

$$\sum_{a=1}^{d} \theta_{a,j} = 1 \quad \text{for all } j = 1, \ldots, w.$$

The full motif model is represented by the matrix

$$\theta = (\theta_1, \ldots, \theta_w),$$

which specifies the probability of each nucleotide at each position. Formally, for a sequence x_i and position j, the probability of observing nucleotide a is

$$P(X_{ij} = a) = \theta_{a,j}, \quad a \in \{A, C, G, T\}.$$

The total probability of the sequence under the motif model is

$$P(x_i \mid \theta) = \prod_{j=1}^{w} \theta_{x_{ij},j}.$$

- **Background model:** The sequence letters are generated independently from a single, position-independent background distribution $\theta_b = (\theta_b^A, \theta_b^C, \theta_b^G, \theta_b^T)$, with

$$P(x_i \mid \theta_b) = \prod_{j=1}^{w} \theta_b^{x_{ij}}.$$

Since we do not observe which model generated each sequence, we introduce a hidden binary variable Z_i for each sequence x_i:

$$Z_i = \begin{cases} 1, & \text{if } x_i \text{ is generated from the motif model,} \\ 0, & \text{if } x_i \text{ is generated from the background model.} \end{cases}$$

We assume the prior probability of motif-origin sequences is known and fixed:

$$P(Z_i = 1) = \alpha, \quad P(Z_i = 0) = 1 - \alpha,$$

where $\alpha \in (0, 1)$.

The goal is to estimate the parameters θ and θ_b based on the observed sequences $\{x_i\}_{i=1}^{k}$, using the EM algorithm to handle the unobserved variables Z_i.

1.2 Likelihood Function

Given k observed DNA sequences $x_1, \ldots, x_k$, where each sequence is of length $w = 3$, and assuming that each sequence was generated either from the motif model or the background model, the marginal likelihood of the data is:

$$L(\Theta) = \prod_{i=1}^{k} \left[\alpha \cdot P(x_i \mid \theta) + (1-\alpha) \cdot P(x_i \mid \theta_b)\right],$$

where $\Theta = (\theta, \theta_b)$ is the full set of model parameters, and α is the known prior probability that a sequence comes from the motif model.

The probability of a sequence x_i under the motif model is:

$$P(x_i \mid \theta) = \prod_{j=1}^{w} \theta_{x_{ij},j},$$

and under the background model:

$$P(x_i \mid \theta_b) = \prod_{j=1}^{w} \theta_{b,x_{ij}},$$

where x_{ij} denotes the nucleotide at position j in sequence x_i.

The goal is to estimate the parameters θ and θ_b by maximizing the log-likelihood:

$$\hat{\Theta} = \arg\max_{\Theta} \log L(\Theta).$$

2 EM Algorithm

To estimate the unknown model parameters in the presence of hidden variables, we use the Expectation-Maximization (EM) algorithm. In our setting, the unobserved variables are the binary indicators $Z = (z_1, \ldots, z_k)$, where each $z_i \in \{0, 1\}$ denotes whether the sequence x_i originates from the motif model ($z_i = 1$) or from the background model ($z_i = 0$).

Since we do not observe these labels directly, we use the EM algorithm to iteratively refine our estimates of the parameters θ and θ_b. Each iteration consists of two steps:

2.1 E-step

In the E-step, we compute the expected value of the hidden variable z_i for each sequence. Let γ_i denote the posterior probability that sequence x_i comes from the motif model:

$$\gamma_i = \mathbb{E}[z_i] = \frac{\alpha \cdot P(x_i \mid \theta)}{\alpha \cdot P(x_i \mid \theta) + (1-\alpha) \cdot P(x_i \mid \theta_b)}.$$

2.2 M-step

In the M-step, we update the parameters based on the soft assignments γ_i computed in the E-step.

Motif model parameters: For each position $j = 1, \ldots, w$ and nucleotide $a \in \{A, C, G, T\}$ (indexed as $a = 1, \ldots, d$), we update the motif distribution as:

$$\theta_{a,j} = \frac{\sum_{i=1}^{k} \gamma_i \cdot \mathbb{I}[x_{ij} = a]}{\sum_{i=1}^{k} \gamma_i},$$

where $\mathbb{I}[x_{ij} = a]$ is an indicator function that is 1 if the j-th letter of sequence x_i is a, and 0 otherwise.

Background model parameters: The background distribution is position-independent, so for each letter a we update:

$$\theta_{b,a} = \frac{\sum_{i=1}^{k}(1-\gamma_i)\cdot\sum_{j=1}^{w}\mathbb{I}[x_{ij}=a]}{w\cdot\sum_{i=1}^{k}(1-\gamma_i)}.$$

2.3 Summary

We repeat the E-step and M-step until the change in log-likelihood is below a fixed threshold or a maximum number of iterations is reached. The output is the estimated parameter set $\hat{\Theta} = (\hat{\theta}, \hat{\theta}_b)$ that best explains the observed data under the model.

3 Details of Implementation

Several technical choices were made to improve the stability and accuracy of the EM algorithm implementation:

Parameter Initialization To ensure faster and more stable convergence of the EM algorithm, I chose to use a data-driven initialization instead of random initialization. Randomly initializing the parameters Θ and Θ_B can lead to high variability in results and increase the risk of getting stuck in a local extremum.

In my implementation, the motif parameters Θ are initialized based on the observed nucleotide frequencies at each position in the input sequences. The background distribution Θ_B is initialized using the overall nucleotide frequencies across all sequences. This approach makes use of the information available in the data and more accurately reflects its structure, leading to more effective parameter estimation during the EM iterations.

Log-space computations A common technique to avoid numerical underflow in probabilistic models is to perform calculations in log-space. Instead of multiplying small probabilities directly, it's much safer numerically to add their logarithms. I used this approach in my implementation when computing the likelihoods of sequences under both the motif and background models. In particular, the `np.logaddexp` function was used to compute expressions like $\log(a+b)$ in a more stable way. This helped the algorithm work reliably even with very small values or longer sequences.

Numerical stability A small constant (10^{-12}) is added to all probability values before logarithm calculation to avoid numerical errors related to $\log(0)$.

Stop criteria The algorithm stops when the improvement in total log-likelihood between iterations is smaller than a small threshold (10^{-6}) or when it reaches the maximum number of iterations (the default is 500).

4 Results

I evaluated the model's ability to recover motif parameters from simulated sequence data. To measure estimation accuracy, I compared the inferred and true distributions using a standard similarity metric, described below.

Measure of Distribution Similarity

To evaluate the quality of parameter estimation in our model, we used the Total Variation Distance (TVD) between the true and estimated probability distributions.

Let $p = (p_1, \ldots, p_n)^T$ and $q = (q_1, \ldots, q_n)^T$ be two discrete probability distributions over the set $\{1, \ldots, n\}$. The TVD is defined as:

$$d_{tv}(p, q) = \frac{1}{2} \sum_{i=1}^{n} |p_i - q_i|$$

This value ranges from 0 to 1, where 0 indicates identical distributions and 1 means completely disjoint distributions. TVD measures the maximum difference in assigned probabilities between two distributions.

In our context, we compare two groups of distributions:

- The background distribution θ_b — a vector of nucleotide probabilities outside the motif,
- The motif distribution θ — a $4 \times w$ matrix where each column θ_i describes the nucleotide probabilities at position i in the motif.

To summarize the difference between the original and estimated parameters, we compute the weighted average TVD:

$$d_{tv,avg} = \frac{1}{w+1} \left(d_{tv}(\theta_b^{orig}, \theta_b^{estim}) + \sum_{i=1}^{w} d_{tv}(\theta_i^{orig}, \theta_i^{estim}) \right)$$

This value reflects how accurately the background and each motif position were estimated.

4.1 Testing Initialization Parameters

The model was tested on a simulated dataset with:

- motif length: $w = 3$,
- number of sequences: $k = 100, \quad k = 10$,
- motif fraction: $\alpha = 0.5$.

The motif matrix θ (size $4 \times w$) and the background distribution θ_b were created by sampling from a Dirichlet distribution with parameter 1, so that each column sums to 1 and represents nucleotide probabilities.

Each sequence in X was made by first deciding if it belongs to the motif with probability α. If yes, nucleotides were drawn from θ; otherwise, from θ_b.

This dataset served to evaluate the EM algorithm by comparing how accurately it estimated the parameters when initialized randomly versus with a fixed starting point, using TVD as the measure of similarity.

Results: For both $k = 100$ and $k = 10$ sequences, the TVD values were nearly identical regardless of whether the parameters were initialized randomly or with a fixed starting point. Specifically, for $k = 100$, both models achieved $d_{tv} = 0.0760$, while for $k = 10$, the TVD was 0.0895 in both cases. These results indicate that, for the tested data, the EM algorithm converges to similar parameter estimates independent of the initialization method. The similarity of results regardless of initialization suggests that local minima likely have little effect on parameter estimation for this dataset, allowing the EM algorithm to find good solutions consistently.

4.2 Impact of Sample Size on Estimation Accuracy

To further evaluate the performance of the EM algorithm, an additional experiment was conducted to test how the estimation accuracy depends on the number of sequences k. The motif length was fixed at $w = 3$ and the motif fraction at $\alpha = 0.5$. The true motif matrix θ and background distribution θ_b were again generated from Dirichlet distributions.

For each tested value of $k \in \{10, 50, 100, 250, 500, 1000, 2000\}$:

- A dataset was generated using the true parameters.
- The EM algorithm was run multiple times with random initialization.
- The Total Variation Distance (TVD) between the estimated and true parameters was computed to quantify estimation error.

To reduce random variability, for each k the experiment was repeated 10 times and the average TVD with standard deviation was calculated.

Results: The experiment confirmed that increasing the sample size significantly improves estimation accuracy. For small samples (e.g., $k = 10$), the average TVD was around 0.40 with higher variability. As the sample size increased, the average TVD decreased rapidly: for $k = 1000$ it dropped to about 0.025, and for $k = 2000$ it reached as low as 0.019. This demonstrates that more data allow the EM algorithm to recover the motif and background parameters more precisely. The decreasing standard deviation also indicates greater consistency in estimates for larger datasets.

The relationship between sample size and estimation error is summarized in Figure 1, which shows the average TVD with error bars for each tested k.

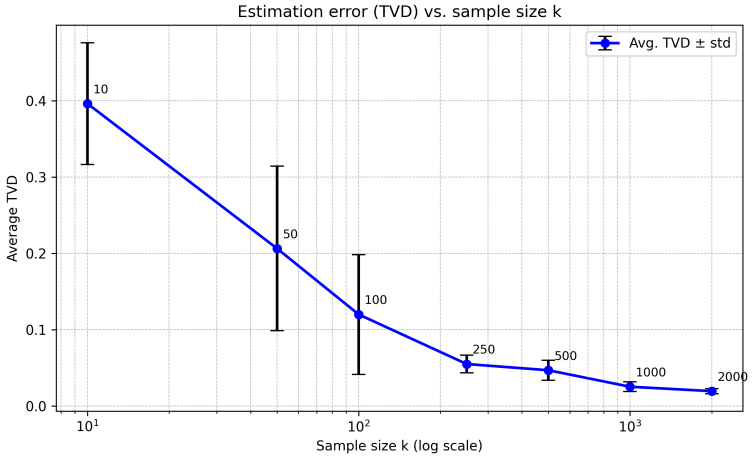


Figure 1: Average Total Variation Distance (TVD) on sample size k with standard deviation error bars.

These results highlight the importance of sufficient data for reliable motif discovery using probabilistic models.